Rounding to a Given Place Value

Exam code: 0607

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Updated on 19 January 2026

Rounding to a given place value

How do I round a number to a given place value?

- Identify the **digit** in the **required place value**
- Identify the **two options** that the number could round to
  - **Count in the units you are rounding to**,
  - Find the first value below and the next value above the number you are rounding
    - E.g. Round 1294 to the nearest 100, count in 100's
    - The number will round down to 1200 or up to 1300
  - Be careful if your digit is a **9** and the next number up will **affect the higher place values**
    - E.g. Rounding 1798 to the nearest 10, count in 10's
    - The number will round down to 1790 or up to 1800
- **Circle the number to the right** of the required place value
  - If the circled number is **5 or more** then you round to the **bigger number**
  - If the circled number is **less than 5** then you round to the **smaller number**
  - Put a **zero** in any **following place values before the decimal point**
    - E.g. 1567.45 to the nearest 100 would be 16**00**

How do I round a number to a given decimal place?

- Identify the **position** of the **decimal place** you are rounding to
- Identify the t**wo options** that the number could round to
  - E.g. Round 7.82741 to 3 d.p., count in 0.001's
  - The number will round down to 7.827 or up to 7.828
- **Circle the number to the right** of the required decimal place
  - If the circled number is **5 or more** then you round to the **bigger number**
  - If the circled number is **less than 5** then you round to the **smaller number**
    - E.g. 2.435123 to the nearest 2 d.p. would be 2.44

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

place you are rounding to
    - E.g. 1267 to the nearest 100 is 13**00**
    - But 1.267 to two decimal places (nearest 100th) is 1.27 not 1.270
  - If asked for a **certain number of decimal places**, you must give an answer with that number of decimal places
    - E.g. 2.395 to two decimal places is 2.40 (do not write 2.4)

Worked Example

Round the following numbers to 2 decimal places.

(i) 345.254

(ii) 0.295 631

(iii) 4.998

Answer:

(i) Identify the second decimal place (5)
Count in 0.01's, identify the first number below (345.25) and the next number above (345.26)
Circle the digit to the right of the second decimal place (4)

$$345.25\,4$$

As this digit is less than 5 we will round the number down

345.25 (2 d.p.)

(ii) Identify the second decimal place (9)
Count in 0.01's, identify the first number below (0.29) and the next number above (0.30)
Circle the digit to the right of the second decimal place (5)

$$0.29\,5\,631$$

As this digit is greater than or equal to 5 we will round the number up

- The zero is important to show we have rounded to two decimal places

0.30 (2 d.p.)

(iii) Identify the second decimal place (9)
Count in 0.01's, identify the first number below (4.99) and the next number above (5.00)
Circle the digit to the right of the second decimal place (8)

$$4.99\,8$$

- Two zeros are needed to show we have rounded to 2 decimal places

5.00 (2 d.p.)

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